

A hybrid fix-and-optimize and simulated annealing approaches for nurse rostering problem

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ARTICLE INFO

Keywords:

OR in health services
Nurse rostering problem
Fix-and-relax
Fix-and-optimize
Simulated annealing

ABSTRACT

The Nurse Rostering Problem (NRP) is a complex scheduling problem in which nurses must be assigned to shifts according to a set of constraints. The NRP is difficult to solve to optimality due to its combinatorial structure. In this paper, we propose a hybrid solution algorithm that combines Mixed Integer Programming (MIP) based heuristics—namely Fix-and-Relax (F&R) and Fix-and-Optimize (F&O)—and Simulated Annealing (SA) to solve the NRP. In MIP-based heuristics, a problem is decomposed into a set of smaller problems and each problem is iteratively solved to optimality. In the hybrid approach, high-quality initial solutions are obtained via the F&R algorithm and fed to the SA part of the algorithm. When solutions can no longer be improved, the F&O algorithm is inserted into the SA algorithm. This enables the hybrid algorithm to diversify the search space and provide better solutions. To assess the quality and efficiency of the hybrid approach, we use 24 publicly available test instances recently introduced in literature in the field. Computational results show that the hybrid method outperforms other advanced solution techniques in most of the test data and seven new best-known results are reported.

1. Introduction

Personnel scheduling problems have been studied by the Operations Research (OR) community since the 1950s. Scheduling today is notably different than it was that time, as many new features have been introduced to the process (Vanden Bergh et al., 2013). For example, employee satisfaction has become an important part of the scheduling effort. Employees are offered part-time and full-time opportunities as well as flexible working hours. Their preferences (i.e. desired shifts, days-off, planned vacations) are taken into consideration when a work schedule is developed.

According to Baker (1976), personnel scheduling problems are generally classified in three groups: time-of-day scheduling, day-of-week scheduling, and a combination of both. In time-of-day scheduling, shift start and end times are scheduled in a daily planning horizon. Weekly planning horizon is used in day-of-week scheduling in which a facility's operating week may not match employees' working week. Typical scenario is when employees work five days a week while a facility operates seven days a week. In the last group, scheduling must be done on both a daily and weekly basis. Hospitals are an example of this combined group. They must operate at all times, and employees have a variety of shifts in a day.

In hospitals, nurses are one of many scarce resources. Hospitals produce work schedules that ensure nurses are on staff 24 h a day, seven days a week. The quality of nurse schedules affects the quality of health care (Oldenkamp, 1996), but scheduling is generally done manually. Head nurses or hospital scheduling departments spend significant time constructing schedules that satisfy many constraints. Factors such as preset shifts, nurse requests, and last minute changes make the task tedious and time consuming (Cheang, Li, Lim, & Rodrigues, 2003). Therefore, nurse scheduling has attracted the attention of the OR community, which has developed the Nurse Rostering Problem (NRP)—also known as the Nurse Scheduling Problem (NSP)—to address these concerns.

The NRP generally has two types of constraints. Hard constraints (HC) are those that must be satisfied in order to generate feasible rosters. Soft constraints (SC) are not necessary conditions for feasible schedules, but violating them causes a penalty. For example, one HC is that a nurse cannot be assigned to a morning shift directly following a late shift because there must be rest time after a shift. An example of an SC would be a nurse asking to be assigned to a certain shift on a certain day. If this request is not fulfilled, a feasible roster can still be created, although it would result in a penalty because of the nurses's dissatisfaction.

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<https://doi.org/10.1016/j.cie.2020.106531>

Received 29 March 2019; Received in revised form 9 November 2019; Accepted 4 May 2020

Available online 16 May 2020

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In the literature, solution techniques for the NRP can be divided into three categories: exact methods, heuristics, and hybrid solutions. The exact methods are integer programming (IP) and constraint programming (CP). These methods guarantee optimality, but computational times increase drastically as problem sizes increase. The heuristics category can provide high-quality solutions in faster processing times, but the solutions may not be optimal. Heuristics include many solution approaches such as variable neighborhood search (VNS), tabu search (TS), simulated annealing (SA), genetic algorithm (GA), ant colony optimization (ACO), electromagnetic algorithm (EM), scatter search (SS), memetic algorithm (MA), tailor-made heuristics, estimation of distribution algorithms (EDA), and case-based reasoning (CBR). The final category of solutions techniques, hybrid solutions, is relatively a new area of study. It combines different solution techniques to achieve greater strength and flexibility. For applications of several metaheuristic approaches, interested readers may refer to recent papers by Hamza, Abderazek, Lakhdar, Ferhat, and Yıldız (2018), and Yıldız, Abderazek, and Mirjalili (2019).

Applications of quantitative models in NRP date back to the 1970s and have been addressed widely in the literature. An interested reader may also refer to Ernst, Jiang, Krishnamoorthy, and Sier (2004), Vanden Bergh et al. (2013), and De Bruecker, Van den Bergh, Beliën, and Demeulemeester (2015) for a detailed bibliography of personnel rostering publications; Cheang et al. (2003), Burke, De Causmaecker, Berghe, and Van Landeghem (2004), De Causmaecker and Vanden Berghe (2011) for the detailed nurse rostering literature. Besides nurses, the scheduling of other types of medical personnel like physicians have been subject to many research studies. In a recent survey paper, Erhard, Schoenfelder, Fügner, and Brunner (2018) provide a review on physician scheduling problem. Very recently, Kletzander and Musliu (2020) propose a new framework for general employee scheduling that allows independent handling of various constraints in a unified way.

Table 1 represents a comprehensive summary of studies in the NRP field in the last several decades.

Two of these studies are particularly notable because they use similar NRP models and experiment on the same data sets as in our work. Curtois and Qu (2014) use ejection chain (EC) and branch and price (B&P) algorithms, and the Gurobi optimizer to solve the test data. Their results indicate that the B&P method is effective on smaller instances but it is inadequate on larger test data, and it runs out of memory at times. Conversely, the EC metaheuristic finds good solutions on the larger data sets but is outperformed by the B&P method on smaller sets. The second study experimenting on the same instances as our work is by Rahimian, Akartunali, and Levine (2017b), who hybridize the VNS algorithm using IP to solve the NRP. In the study, initial solutions are generated using a greedy heuristic and improved using the hybrid method. Five different neighborhoods are applied to improve schedules during the VNS phase and an IP based ruin-and-recreate framework is embedded into the process to further improve solutions. The study reports new best-known results and compares findings with studies in the literature. It also generates new test data and makes the instances publicly available for other researchers for benchmarking.

Most of the studies in the literature utilize stand-alone heuristic approaches such as (Burke et al., 1999; Aickelin & Dowsland, 2000; Burke, Curtis, Post, Qu, & Veltman, 2008; Awadallah, Bolaji, & Al-Betar, 2015) or exact solution techniques such as (Azaiez & Al Sharif, 2005; Burke & Curtois, 2014; Dowsland & Thompson, 2000; Maenhout & Vanhoucke, 2010). There are still only a handful of publications (Burke, Curtois et al., 2010; Qu & He, 2008; Rahimian et al., 2017a, 2017b) that investigate the combination of the both approaches. Moreover, studies applying hybrid approaches generally use VNS for the heuristic part. Our study proposes a hybrid solution methodology that combines MIP based heuristics with SA.

In our study, we propose a hybrid approach to solve the NRP. Our approach integrates Mixed Integer Programming (MIP)-based Fix-and-

Table 1
Summary of the NRP research.

Study	Solution category/approach
Warner and Prawda (1972)	Exact/IP
Brusco and Jacobs (1993)	Heuristics/SA
Darmoni et al. (1995)	Exact/CP
Weil, Heus, Francois, and Poujade (1995)	Exact/CP
Brusco and Jacobs (1995)	Heuristics/SA
Berrada, Ferland, and Michelon (1996)	Exact/IP
Beaumont (1997)	Exact/IP
Dowsland (1998)	Heuristics/TS
Burke, De Causmaecker, and Vanden Berghe (1999)	Heuristics/TS
Easton and Mansour (1999)	Heuristics/GA
Dowsland and Thompson (2000)	Exact/IP
Meyer auf'm Hofe (2000)	Exact/CP
Aickelin and Dowsland (2000)	Heuristics/GA
Cai and Li (2000)	Heuristics/GA
Kawanaka, Yamamoto, Yoshikawa, Shinigi, and Tsuruoka (2001)	Heuristics/GA
Burke, Cowling, De Causmaecker, and Vanden Berghe (2001)	Heuristics/MA
Ikegami and Niwa (2003)	Exact/IP
Bourdais, Galinier, and Pesant (2003)	Exact/CP
Burke, De Causmaecker, Petrovic, and Vanden Berghe (2003)	Heuristics/VNS
Aickelin and Dowsland (2004)	Heuristics/GA
Bard and Purnomo (2005)	Exact/IP
Azaiez and Al Sharif (2005)	Exact/IP
Cipriano, Gaspero, and Dovier (2006)	Exact/CP
Gutjahr and Rauner (2007)	Heuristics/ACO
Maenhout and Vanhoucke (2007)	Heuristics/EM
Aickelin and Li (2007)	Heuristics/EDA
Beddoe and Petrovic (2007)	Heuristics/CBR
Burke et al. (2008)	Heuristics/VNS
Qu and He (2008)	Hybrid/CP and VNS
Tsai and Li (2009)	Heuristics/GA
Glass and Knight (2010)	Exact/IP
Maenhout and Vanhoucke (2010)	Exact/IP
Burke, Curtois et al. (2010)	Heuristics/SS
Burke, Li et al. (2010)	Hybrid/IP and VNS
Girbea, Suciu, and Sisak (2011)	Exact/CP
Stølevik et al. (2011)	Heuristics/VNS
Bilgin, De Causmaecker, Rossie, and Berghe (2012)	Heuristics/VNS
Lu and Hao (2012)	Heuristics/Tailor-made
Valouxis, Gogos, Goulas, Alefragis, and Housos (2012)	Heuristics/Tailor-made
M'Hallah and Alkhabbaz (2013)	Exact/IP
Soto, Crawford, Bertrand, and Monfroy (2013)	Exact/CP
Curtois and Qu (2014)	Heuristics/EC
Della Croce and Salassa (2014)	Mathheuristics/VNS
Burke and Curtois (2014)	Exact/IP
Tassopoulos, Solos, and Beligiannis (2015)	Heuristics/VNS
Awadallah et al. (2015)	Heuristics/ACO
Santos, Toffolo, Gomes, and Ribas (2016)	Exact/IP
Rahimian, Akartunali, and Levine (2017a)	Hybrid/IP and CP
Rahimian et al. (2017b)	Hybrid/IP and VNS
Liu, Liu, Zhu, Shen, and Dong (2018)	Heuristics/SA
Vermuyten, Rosa, Marques, Beliën, and Barbosa-Póvoa (2018)	Heuristics/VNS
The Proposed Study	Hybrid/IP and SA

Relax (F&R) and Fix-and-Optimize (F&O) heuristics with Simulated Annealing (SA). In MIP-based heuristics, a problem is decomposed into a set of sub-problems, and then each sub-problem is optimized. This process continues iteratively until all the sub-problems are solved. In our proposed NRP implementation framework, we use F&R heuristic as a starting point to find high-quality initial solutions. The NRP is decomposed based on the number of nurses and weeks. The initial solution obtained using the F&R heuristic is then used in the SA part of the framework. Many neighborhood structures are applied during the subsequent iterations to improve the initial solution. When solutions can no longer be improved, the F&O heuristic is injected into the

process as a vehicle to diversify the search space. Low cost days are fixed to their existing values and other days are optimized. This interaction between the SA and F&O algorithms often results in better solutions and leads to intensification. Even when the F&O heuristic produces worse solutions, the diversification of the search space improves the performance of the SA algorithm and the neighborhoods. Results are reported when a termination criterion is met in the SA algorithm. Finally, we use data instances previously reported in the literature to assess the performance of the proposed solution and compare our results to other solution techniques.

The main contributions of the paper to the literature are two-fold: our study is the first implementation of the both MIP-based heuristics to the NRP problem and computational results outperform some of the state-of-the-art solution techniques studied in the literature.

The remainder of this paper is organized as follows. In Section 2, the description, notation, and the mathematical programming formulation of the NRP are defined. Section 3 describes the proposed algorithm and illustrates the implementation with an example. In Section 4, the computational experiments are presented and compared to results previously reported in the literature. Finally, in Section 5, conclusions are drawn and possible future research opportunities are discussed.

2. Problem description

Before describing the problem in detail, some of the useful terms and essential definitions of the NRP literature are provided in the following paragraphs.

Nurse: Person who has completed a program of nursing education and is authorized to care for patients.

Shift: Time period during which nurses perform their duties. There can be multiple shifts in a day (i.e. early shift, day shift, night shift). Shifts have pre-defined start and end times.

Cover: Minimum number of nurses per shift needed to provide sufficient health care services.

Schedule: Work plan that defines assigned shifts for nurses for each day in a timetable.

Planning horizon: Length of one schedule. Each planning horizon is defined by its start and end dates.

Contract: Employment agreement between a nurse and the management of a hospital. Contracts are defined in part by laws and regulations. For example, there is a mandated minimum period of rest following a shift. Thus, hospitals cannot assign a day shift to a nurse directly after a night shift. The law also defines the maximum number of consecutive days that nurses can work without time off as well as their minimum amount of time off. Generally, nurses work five days consecutively and receive at least two days off. Contracts also specify the maximum and minimum number of hours nurses must work in a planning horizon.

Request: Appeal from a nurse that could result in a schedule change. For instance, a nurse may request to be assigned to a morning shift instead of a night shift on a given day.

Fixed assignment: Assignment that is pre-defined before a schedule is generated. Vacations are considered in this group.

The NRP automatically assigns nurses to shifts according to the cover requirements, contractual agreements, nurses' requests, and fixed assignments. Contract requirements and fixed assignments are generally considered HCs, and therefore must be met. In the NRP, cover requirements and requests are SCs; violating them results in costs in the objective function of the problem. The quality of a schedule increases as the number of satisfied SCs increases.

A general formulation of the NRP problem studied in this paper is originally provided by [Curtois and Qu \(2014\)](#). A detailed description is presented in this section to make the paper self-contained and easy to follow.

Table 2

Notation of the model.

Sets	Definition
I	Set of nurses ($i = 1, 2, \dots, I$)
h	Number of days in the planning horizon
D	Set of days in the planning horizon = $\{1 \dots h\}$ ($d = 1, 2, \dots, D$)
W	Set of weekends in the planning horizon = $\{1 \dots h/7\}$ ($w = 1, 2, \dots, W$)
T	Set of shift types ($t = 1, 2, \dots, T$)
R_t	Set of shift types that cannot be assigned immediately after shift type t
N_i	Set of days that nurse i cannot be assigned a shift on
l_t	Length of shift type t in minutes
s	Number of restricted days in a pattern
Parameters	Definition
m_{it}^{max}	Maximum number of shifts of type t that can be assigned to nurse i
b_i^{min}	Minimum number of minutes that nurse i must be assigned
b_i^{max}	Maximum number of minutes that nurse i can be assigned
c_i^{min}	Minimum number of consecutive shifts that nurse i must work
c_i^{max}	Maximum number of consecutive shifts that nurse i can work
o_i^{min}	Minimum number of consecutive days off that nurse i can be assigned
a_i^{max}	Maximum number of weekends that nurse i can work
q_{idt}	Penalty if shift type t is not assigned to nurse i on day d
p_{idt}	Penalty if shift type t is assigned to nurse i on day d
u_{dt}	Preferred total number of nurses assigned shift type t on day d
v_{dt}^{min}	Penalty weight when below the preferred cover for shift type t on day d
v_{dt}^{max}	Penalty weight when exceeding the preferred cover for shift type t on day d
Decision Variables	Definition
x_{idt}	1 if nurse i is assigned shift type t on day d , 0 otherwise
k_{iw}	1 if nurse i works on weekend w , 0 otherwise
y_{dt}	Total below the preferred cover for shift type t on day d
z_{dt}	Total above the preferred cover for shift type t on day d

2.1. Notation

Table 2 describes the notation of the model.

2.2. Constraints

The HCs of the mathematical model are as follows:

$$\sum_{t \in T} x_{idt} \leq 1, \quad \forall i \in I, d \in D \quad (1)$$

$$x_{idt} + x_{i(d+1)u} \leq 1, \quad \forall i \in I, d \in \{1 \dots h-1\}, t \in T, u \in R_t \quad (2)$$

$$\sum_{d \in D} x_{idt} \leq m_{it}^{max}, \quad \forall i \in I, t \in T \quad (3)$$

$$b_i^{min} \leq \sum_{d \in D} \sum_{t \in T} l_t x_{idt} \leq b_i^{max}, \quad \forall i \in I \quad (4)$$

$$\sum_{j=d}^{d+c_i^{max}} \sum_{t \in T} x_{ijt} \leq c_i^{max}, \quad \forall i \in I, d \in \{1 \dots h - c_i^{max}\} \quad (5)$$

$$\sum_{t \in T} x_{idt} + \left(s - \sum_{j=d+1}^{d+s} \sum_{t \in T} x_{ijt} \right) + \sum_{t \in T} x_{i(d+s+1)t} > 0, \quad \forall i \in I, s \in \{1 \dots c_i^{min}\}, d \in \{1 \dots h - (s+1)\} \quad (6)$$

$$\left(1 - \sum_{i \in T} x_{idt}\right) + \sum_{j=d+1}^{d+s} \sum_{i \in T} x_{ijt} + \left(1 - \sum_{i \in T} x_{i(d+s+1)t}\right) > 0, \\ \forall i \in I, s \in \{1 \dots o_i^{\min} - 1\}, d \in \{1 \dots h - (s + 1)\} \quad (7)$$

$$k_{iw} \leq \sum_{i \in T} x_{i(7w-1)t} + \sum_{i \in T} x_{i(7w)t} \leq 2k_{iw}, \quad \forall i \in I, w \in W \quad (8)$$

$$\sum_{w \in W} k_{iw} \leq a_i^{\max}, \quad \forall i \in I \quad (9)$$

$$x_{idt} = 0, \quad \forall i \in I, d \in N_i, t \in T \quad (10)$$

$$\sum_{i \in I} x_{idt} - z_{dt} + y_{dt} = u_{dt}, \quad \forall d \in D, t \in T \quad (11)$$

Constraints (1) ensure that a maximum of one shift per day can be assigned to each nurse. Because of minimum rest-time regulations, certain shifts cannot follow others. As mentioned earlier, an early shift assignment cannot follow a late shift. This requirement is satisfied by Constraints (2).

Constraints (3) limit the number of shift assignments. The total number of assignments of a shift in the entire planning horizon must be less than or equal to its maximum allowed limit. Constraints (4) control total minutes worked in a planning horizon, which must be within the thresholds defined in the contracts. There can be differences in time limits based on nurses' employment statuses. For example, part-time nurses do not work as much as full-time nurses.

Constraints (5) enforce the maximum number of consecutive shifts a nurse can be assigned. When the limit is reached, a day off must be granted for the next day. Hospitals also want to ensure that the minimum number of consecutive work shifts is achieved. Therefore, when nurses are assigned to shifts, they should be scheduled for successive days. Constraints (6) ensure that the total number of work days is greater than this pre-defined requirement. These constraints are modeled by preventing every pattern below the minimum provision. For example, if a nurse must work at least three days before taking a day off, "off-on-off" and "off-on-off-off" patterns cannot be allowed. The same approach is utilized for modeling Constraints (7), which require that each nurse receives the minimum number of days off after working consecutively. If a nurse must take off for a minimum of three days, these constraints will prevent "on-off-on" and "on-off-off-on" patterns.

Because hospitals operate continuously, having adequate staff on weekends is a critical consideration. Constraints (8) and (9) ensure the maximum number of weekends nurses work to be less than the limit defined in their contracts. Any Saturday or Sunday shift is defined weekend work. Vacations are considered in Constraints (10). A nurse cannot be assigned a shift while on vacation.

Finally, Constraints (11) relate y and z variables to x to identify the gap between the number of nurses assigned and daily cover requirements.

2.3. Objective function

$$\sum_{i \in I} \sum_{d \in D} \sum_{t \in T} q_{idt} (1 - x_{idt}) + \sum_{i \in I} \sum_{d \in D} \sum_{t \in T} p_{idt} x_{idt} + \sum_{d \in D} \sum_{t \in T} v_{dt}^{\min} y_{dt} + \sum_{d \in D} \sum_{t \in T} v_{dt}^{\max} z_{dt}$$

The first two parts of the objective function is considered under SC1. The first expression of the objective function deals with nurses' shift-on requests. The second one addresses shift off requests. SC1 considers scenarios in which a nurse asks to be assigned to or excused from a certain shift on a certain day. The third part of the objective function (SC2) penalizes the number of staff below the needed number of nurses for a day and for a shift. The last part (SC3) accounts for over staffing for a given shift on a given day. While overstaffing (SC3) is not an optimal situation, understaffing (SC2) causes more significant problems

Table 3
Weights of the constraints.

Constraint	Description	Weight
SC1	Shift on/off requests	1 – 3
SC2	Understaffing	100
SC3	Overstaffing	1

Table 4
Notation of the algorithms.

Notation	Definition
P	Set of sub-problems
P _{WD} , P _{DD} , P _{ND}	Set of sub-problems with week, day, and nurse decomposition
x _{idt} ^O	Set of x variables that are optimized
x _{idt} ^T	Set of x variables that are to be optimized in future iterations
x _{idt} ^F	Set of x variables that are fixed
x _{idt} ^R	Set of x variables that are relaxed continuously
S	Solution
S'	Current solution
S*	Best solution
T	Temperature
T ₀	Initial temperature
T _{min}	Final temperature
β	Cooling rate
N _k	Neighborhood k
I _{max}	Number of maximum iterations
J _{max}	Number of maximum iterations allowed without an improvement

Algorithm 1: Fix-and-Relax

Input: Problem instance, week decomposition

Output: Status, Feasible solution if found

```

1 : PWD ← Decompose(P);
2 : RelaxVariables(xidt);
3 : foreach p in PWD do
4 :   Clear(xidtT);
5 :   (status, solution) ← Solve(p);
6 :   if status == notFeasible then
7 :     return(status, 0);
8 :   end if
9 :   xidtF ← Update(xidtO);
10 : end
11 : return(Feasible, solution);

```

Fig. 1. Pseudo-code of the F&R heuristic.

in health care coverage. Therefore, the cost of understaffing weighs more than any other cost in the scheduling process.

Table 3 presents the weights of the SCs.

3. Solution methodology

This paper proposes a hybrid solution methodology to solve the NRP. It is a combination of the MIP based F&R and F&O heuristics and SA. The F&R heuristic, which is first introduced by Dillenberger, Escudero, Wollensak, and Zhang (1994), finds quick, high-quality initial solutions. During this algorithm, the problem is decomposed into a

Algorithm 2: Simulated Annealing**Input:** Initial solution**Output:** Final solution

```

1 : Initialize  $T_0, T_{\min}, \beta, I$ ;
2 :  $T = T_0$ ;  $S^* = S$ ;  $j = 0$ ;
3 : while  $T_{\min} < T$  do
4 :    $i = 0$ ;
5 :   while  $i < I_{\max}$  do
6 :      $k = \text{Random}()$ ;
7 :      $S' = \text{Generate}(S, N_k)$ ;
8 :     if  $\text{Cost}(S') < \text{Cost}(S)$  then
9 :        $S = S'$ ;
10 :    else
11 :       $\Delta = \text{Cost}(S') - \text{Cost}(S)$ ;
12 :      if  $\text{Random}(0,1) < \exp(-1*(\Delta/T))$  then
13 :         $S = S'$ ;
14 :      end if
15 :    end if
16 :    if  $S' < S^*$  then
17 :       $S^* = S'$ ;
18 :    else
19 :       $j = j + 1$ ;
20 :    end if
21 :     $i = i + 1$ ;
22 :    if  $j < J_{\max}$ 
23 :       $S = \text{FixAndOptimize}(S)$ ;
24 :       $j = 0$ ;
25 :    end if
26 :  end while
27 :   $T = T * \beta$ ;
28 : end while
29 : return  $S^*$ ;

```

Fig. 2. Pseudo-code of the SA algorithm.

set of smaller sub-problems. The method of decomposition is based on the size of the problem. For smaller problem sizes, we choose to decompose by weeks (WD). For the larger instances, we decompose by nurses (ND). Table 4 provides the notation used to describe the algorithms.

A pseudo-code of the F&R algorithm using week decomposition is described in Fig. 1. The algorithm first uses the Decompose() function to obtain a set of sub-problems based on a chosen decomposition input. The RelaxVariables() block relaxes entire $\times$ variables continuously. For every sub-problem in the set, the algorithm first clears all the fixations

Algorithm 3: Fix-and-Optimize**Input:** Initial solution, day decomposition**Output:** Final solution

```

1 :  $S^* = S$ ;
2 :  $P_{DD} \leftarrow \text{Decompose}(P)$ ;
3 :  $x_{idt}^F \leftarrow \text{Update}(x_{idt})$ ;
4 : foreach  $p$  in  $P_{DD}$  do
5 :    $\text{Clear}(x_{idt}^T)$ ;
6 :    $(\text{status}, S') \leftarrow \text{Solve}(p)$ ;
7 :   if  $\text{status} == \text{feasible}$  then
8 :     if  $\text{Cost}(S') < \text{Cost}(S)$  then
9 :        $S^* = S'$ ;
10 :       $x_{idt}^F \leftarrow \text{Update}(x_{idt}^O)$ ;
11 :    end if
12 :   else
13 :      $\text{Restore}(x_{idt}^T)$ ;
14 :   end if
15 : end
16 : return  $S^*$ ;

```

Fig. 3. Pseudo-code of the F&O heuristic.

and relaxations for the variables ready to be optimized. Then, the problem is solved with an IP solver. If a feasible solution is found, optimized variables are fixed before the next iteration starts in the Update() function. This iterative process continues until all the sub-problems in the set are solved, and a final feasible solution is reported.

When the final solution is reported from the F&R algorithm, it is treated as an initial solution and is passed to the next part of the algorithm, the SA metaheuristic algorithm, which originates in the physical annealing process (Černý, 1985; Kirkpatrick, Gelatt, & Vecchi, 1983). It has many variants, but the version used in the second part of the hybrid framework uses probabilistic acceptance and geometric cooling. In this SA algorithm, SA parameters are put in place and the starting solution-related parameters are set based on the solution received from the F&R algorithm. At each temperature level, a new solution is generated using different neighborhood structures which are defined by the Generate() function. Neighborhoods are selected randomly. Then, the cost of the new feasible solution is calculated. The new cost is immediately accepted if it is lower than the cost of the current schedule. The new cost is accepted in a probabilistic manner even if it is higher than the current cost. This property of the SA algorithm enables the search to expand to global optima.

When a certain criteria is met or there is no improvement at a temperature level, the current schedule is forwarded to the F&O heuristic of the hybrid algorithm by the FixAndOptimize() block. There are couples of benefits of the F&O algorithm in this context which are worth to mention. First of all, the algorithm works as a way to intensify the structure when it finds a better solution. Secondly, and more

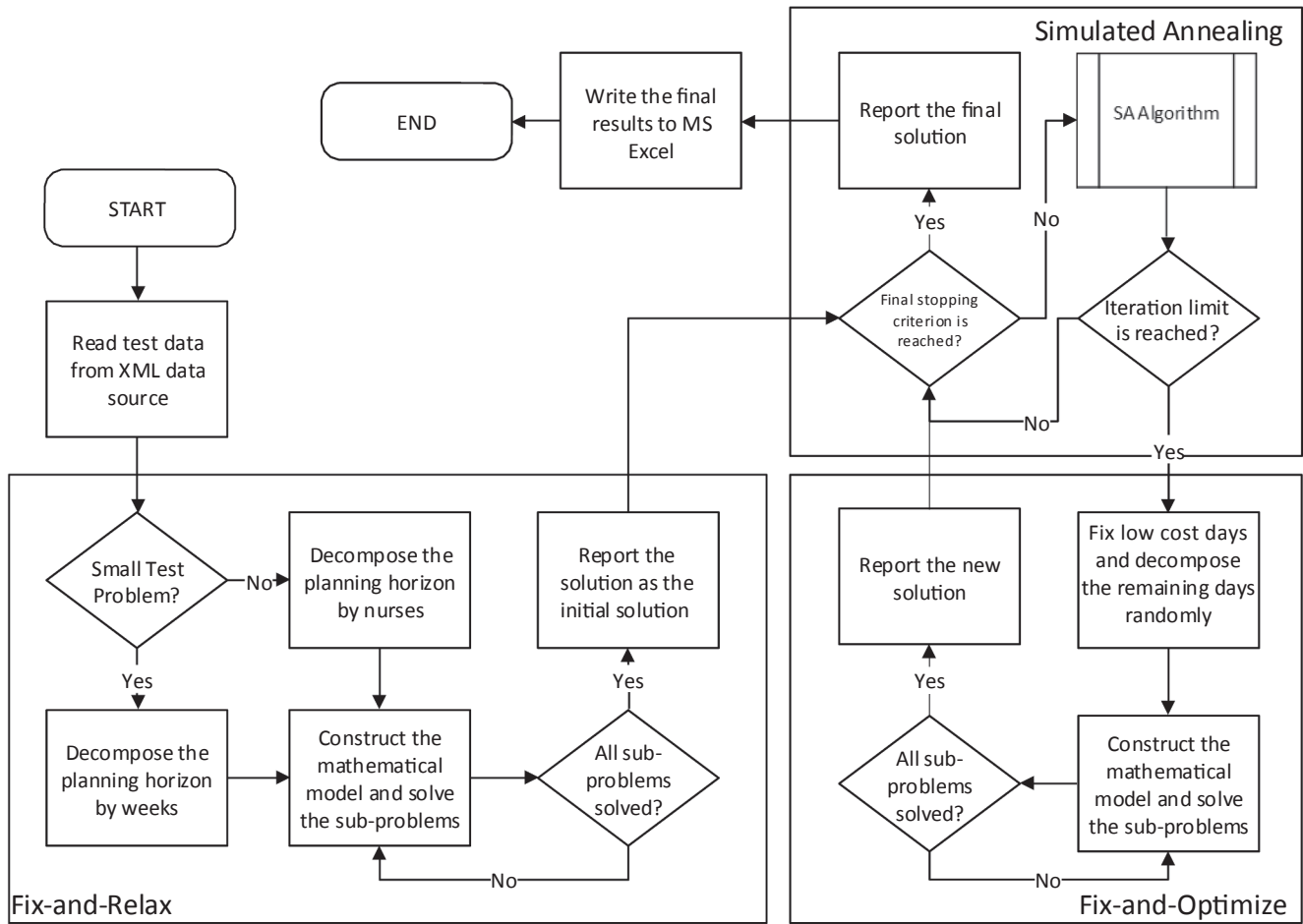


Fig. 4. The overall flow of the hybrid algorithm.

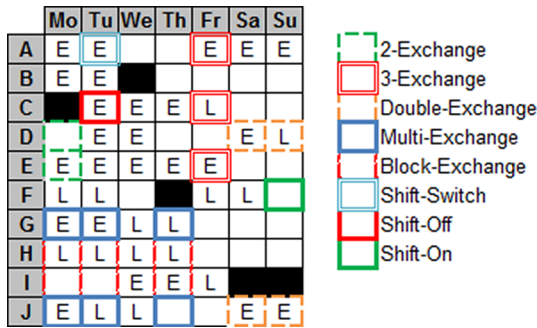


Fig. 5. An illustration of the neighborhoods.

	Mo	Tu	We	Th	Fr	Sa	Su
A	E	E			E	E	E
B	E	E					
C			E	E	E		
D	E	E	E			E	E
E		E	E	E	L		
F	L	L			L	L	
G	E	E	L	L			
H	L	L	L	L			
I			E	E	L		
J	E	E	L	L		E	L

Fig. 6. The new schedule after the neighborhood application.

importantly, the heuristic helps for diversification of the search structure even when no better solution is found due to the parameters of the IP solver, such as reaching the maximum time limit. In any case, diversifying the search space significantly impacts solution quality. Fig. 2 provides a pseudo-code of the SA algorithm. It also details how the SA and F&O algorithms are connected.

The final part of the hybrid algorithm is the F&O heuristic, which is discussed in several studies (Gintner, Kliwer, & Suhl, 2005; Helber & Sahling, 2010; Pochet & Wolsey, 2006; Sahling, Buschkühl, Tempelmeier, & Helber, 2009). Unlike the F&R heuristic, the F&O is an improvement heuristic. An initial solution is provided to the algorithm with a rule of decomposition, and the F&O heuristic improves the initial solution. In the hybrid implementation, the day decomposition (DD) is utilized. First, low-cost days of the solution received from the SA algorithm are identified and fixed to their current values. The rest of the

days in the planning horizon are randomly selected. However, the selection is not entirely random: it accounts for the costs of days by using weighted averages. For example, if the cost of a Monday is four times higher than the cost of a Tuesday for a certain week, the random number generator is more likely to pick Monday. When the cost of a new solution is lower than that of the initial solution provided to the algorithm, it is accepted and the optimized variables are added to the set of fixed variables. If the new solution is not feasible, the Restore() function brings the previous feasible decision variables back into effect. Fig. 3 displays a pseudo-code of the F&O heuristic implemented to solve

Table 5

Summary of the test data (Curtois & Qu, 2014).

Instance	Days	Nurses	Shifts	Day-off Requests	Shift-on/off Requests
Instance 01	14	8	1	8	26
Instance 02	14	14	2	14	62
Instance 03	14	20	3	20	64
Instance 04	28	10	2	20	71
Instance 05	28	16	2	32	106
Instance 06	28	18	3	36	135
Instance 07	28	20	3	40	168
Instance 08	28	30	4	60	225
Instance 09	28	36	4	72	232
Instance 10	28	40	5	80	284
Instance 11	28	50	6	100	336
Instance 12	28	60	10	120	422
Instance 13	28	120	18	240	841
Instance 14	42	32	4	128	359
Instance 15	42	45	6	180	490
Instance 16	56	20	3	120	280
Instance17	56	32	4	160	480
Instance18	84	22	3	176	414
Instance19	84	40	5	320	834
Instance20	182	50	6	900	2318
Instance21	182	100	8	1800	4702
Instance22	364	50	10	1800	4638
Instance23	364	100	16	3600	9410
Instance24	364	150	32	5400	13,809

the NRP.

The flow of the hybrid algorithm is summarized in Fig. 4, which shows the relationships and connection points between each technique.

3.1. Neighborhoods

In this sub-section, we describe the neighborhoods used during the neighborhood generation phase of the SA algorithm (Generate()). These neighborhoods are commonly used in the literature (Burke, Li et al., 2010; Stølevik, Nordlander, Riise, & Frøyseth, 2011; Rahimian et al., 2017a, 2017b), apart from the last neighborhood, which is a new structure proposed in this study.

2-Exchange: Shifts between two randomly selected nurses are exchanged on a randomly selected day

3-Exchange: Shifts between three randomly selected nurses are exchanged on a randomly selected day

Double-Exchange: Shifts between two randomly selected nurses are exchanged on two randomly selected consecutive days

Multi-Exchange: Shifts between two randomly selected nurses are exchanged on three to six randomly selected days. Days are not

necessarily consecutive.

Block-Exchange: Shifts between two randomly selected nurses are exchanged on three to six randomly selected consecutive days.

Shift-Switch: An existing shift of a randomly selected nurse on a randomly selected day is switched to another shift.

Shift-Off: An existing shift of a randomly selected nurse on a randomly selected day is removed.

Shift-On: A new shift is assigned to a free slot of a randomly selected nurse on a randomly selected day.

Fig. 5 gives examples of various neighborhood applications. For instance, Nurses A, C, and E exchange their shifts on Friday as part of the 3-Exchange neighborhood. Nurses G and J multi-exchange their shifts on Monday, Tuesday, and Thursday. While Nurse A switches from shift E to shift L on Tuesday, Nurse C receives a shift off on the same day. Although Nurse F did not have any shifts assigned on Sunday initially, the nurse is assigned to shift L as part of the Shift-On neighborhood.

The resulting schedule after the applications of the neighborhoods described above is shown in Fig. 6.

4. Computational experiments

In this section, we present an experimental evaluation for the hybrid algorithm. The purpose of section is to answer the following questions:

- Under what circumstances does the algorithm provide high quality initial and final solutions?
- What parameters do lead to better results?
- How does each part of the hybrid algorithm contribute to the final solution?
- How do results of the algorithm compare to the results of the state-of-the-art methods?

The software of the proposed hybrid algorithm is written in Java and the sub-problems are optimized using the IBM ILOG CPLEX Optimization Studio 12.6 Application Programming Interface with default parameters. The code is executed in the Java Runtime Environment 7 and run on a Windows 7 PC with an Intel Core i3 2.27 GHz processor and 3 GB of RAM.

4.1. Dataset

The proposed algorithm is tested on the data instances that have recently been introduced by Curtois and Qu (2014). Extensive computational experiments have been conducted on this data (Employee shift scheduling benchmark data sets). The instances are of different sizes

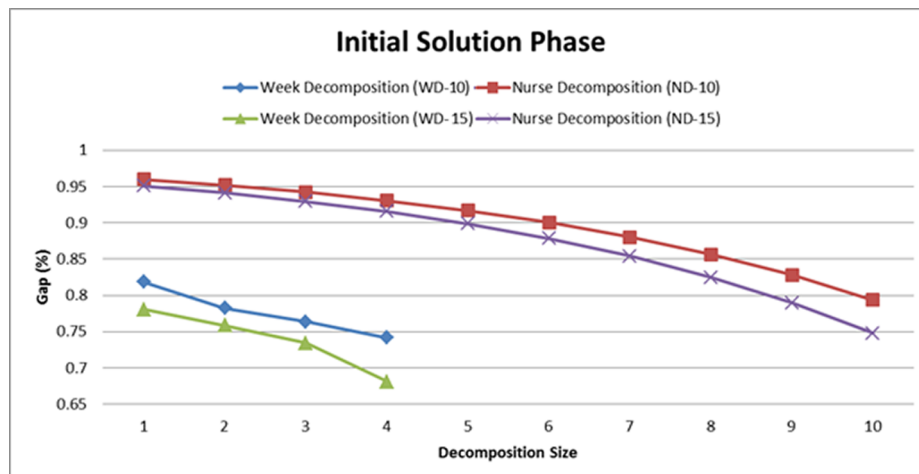


Fig. 7. Performance of the hybrid algorithm on instance 8 over time.

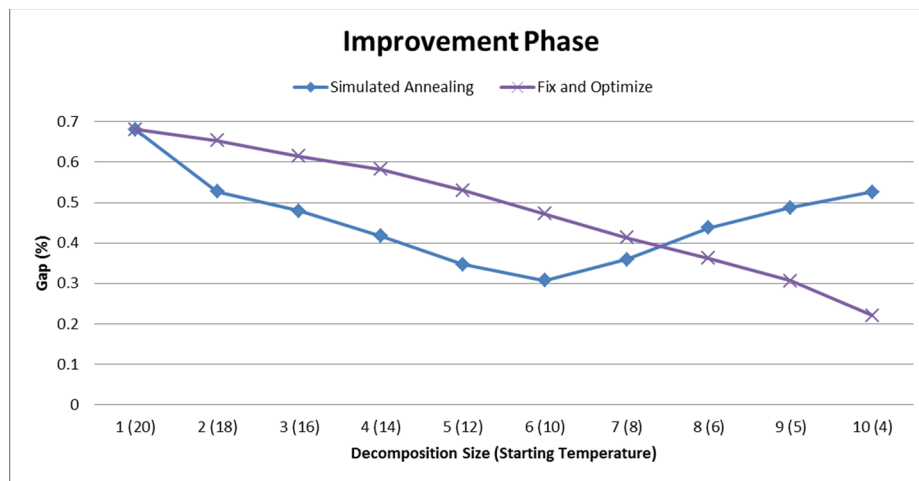


Fig. 8. Performance of the hybrid algorithm on instance 8 over time.

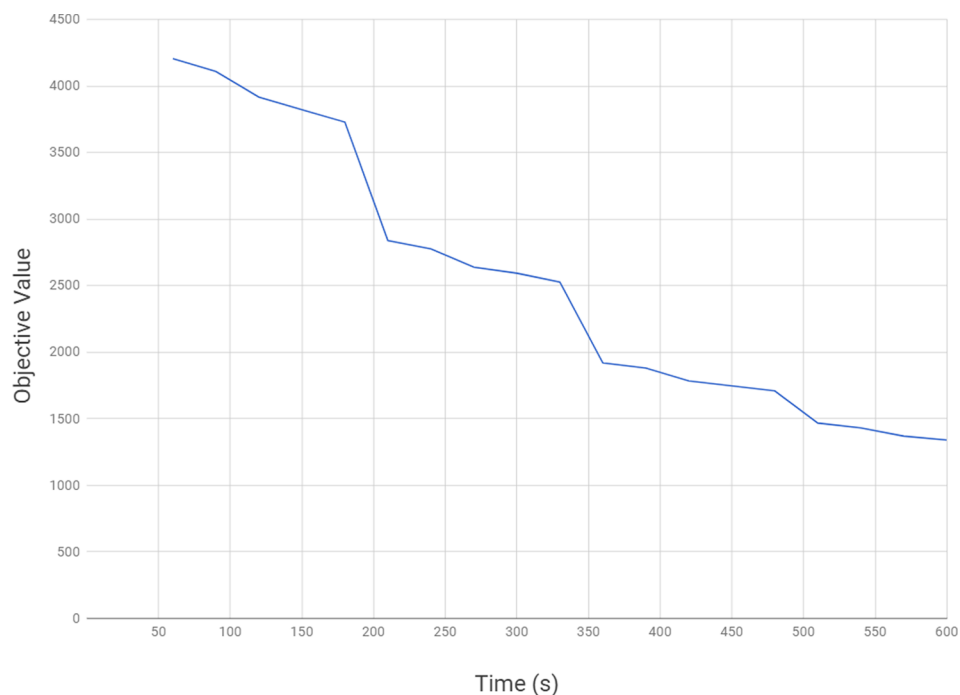


Fig. 9. Performance of the hybrid algorithm on instance 8 over time.

and complexities, which allows a more comprehensive assessment of the efficiency and strength of the proposed solution. Table 5 summarizes the primary characteristics of the instances. There is a direct correlation between the complexity of the data and number of nurse requests.

The planning horizon on instances 20–24 are for six months or a full year. They are not included in our experiments because the data is not practical in real world situations. In reality, the planning horizon of nurse schedules is generally set to be bi-weekly or monthly.

4.2. Parameter setting

This section represents the parameter setting efforts to identify the best setting for the algorithm. Parameters are evaluated separately on both initial solution and improvement phases. Instance 8 is selected for the evaluation process. This instance neither is too small nor too large and it has a good representation of the improvement progress of the overall solution technique. It has 4 week planning horizon and 30

nurses to evaluate. The overall time limit of the entire algorithm is set to 10-minute to be consistent with the literature. 10 percent of the time (60 s) is allocated to the initial solution phase and the rest of the time is used by the SA and F&O algorithms. The algorithm runtime allocation is controlled by the CPLEX solver by limiting optimization of each run. For instance, in order to comply with the 60 s time limit of the F&R algorithm, each week has to be optimized in 15 s at most if the planning horizon is four weeks. The algorithm terminates and returns results when it reaches the iteration time limit. For the initial solution phase where F&R is used, both of the decomposition techniques, WD and ND, are evaluated coupled with the problem time limit (PTL). 10 and 15 s of PTLs are chosen for the parameter setting analysis considering the overall time limit for the initial solution phase to be 60 s.

Fig. 7 represents the results for the initial solution phase. Gap values from the best known result for the Instance 8 are displayed on the Y-axis. X-axis shows different decomposition sizes. For the WD, it represents weeks and for the ND, it displays the percentage of total nurses taken into account during sub-problems. Analyzing this figure, it can be

Table 6
Computational results on 10-minute runtime.

Instance	Initial Solution	% Gap	Final Solution
Instance 01	706	14.02	607
Instance 02	1226	32.46	828
Instance 03	1208	17.14	1001
Instance 04	1931	11.13	1716
Instance 05	1652	30.81	1143
Instance 06	3267	40.31	1950
Instance 07	2070	48.99	1056
Instance 08	4208	68.13	1341
Instance 09	2750	84.04	439
Instance 10	22,066	79.01	4631
Instance 11	27,188	87.34	3443
Instance 12	26,960	85.00	4044
Instance 13	35,560	91.00	3200
Instance 14	23,988	94.60	1295
Instance 15	34,988	87.37	4420
Instance 16	15,809	79.42	3253
Instance 17	34,388	82.15	6138
Instance 18	28,731	82.60	5000
Instance 19	49,661	92.33	3809

observed that WD results with different PTLs are better than the ones in ND. This is expected because the biggest cost component of the problem is the cover penalty. For the ND to provide good results, it needs to include more nurses into sub-problems. As the number of nurses increases for the ND scenario, it starts to generate better results as seen in the figure. But time limits are not adequate to achieve better gap values.

WD-15 with decomposition size 4 generates the best overall solution for the initial solution phase. This schedule is used as a starting schedule for the improve phase as mentioned earlier.

SA and F&O algorithms are evaluated and reported separately for the parameter setting analysis of the improve phase. Stand-alone performances are reported in the Fig. 8. For the SA algorithm, different starting temperatures are used and final solutions are reported. Starting temperature 10 provides the best result. For the F&O algorithm, only DD is evaluated with different number of days to include in the optimization. As expected, increase in number of days in the optimization improves the quality of the solution.

Although F&O reports better solutions in the stand-alone testing, it can only achieve solutions around 20% gap values. The combination of

the SA & F&O provides better final solutions.

The performance of each algorithm in the hybrid solution on Instance 8 is shown in Fig. 9. The F&R heuristic provides an initial solution of 4208 in 60 s. Then, objective values are reported every 30 s until the runtime limit. Lines representing sharp cost declines on the graph are generally as a result of the F&O heuristic. The SA algorithm often results in continuous and steady improvements.

4.3. Results

Computational results for a 10-minute runtime are presented in Table 6. The Initial Solution column shows the results of the F&R heuristic phase, and the Final Solution column shows the results of the hybrid method.

Gap values as shown in Table 6 are relatively smaller for Instances 1–8, where week decomposition is mainly used in the F&R heuristic. This enables the SA part of the algorithm to start from a very high-quality initial solution. Thus, it quickly reaches optimality. As problem sizes increase, the week decomposition often fails to provide a feasible solution. Instances 9–19 are more complex and use large data sets and planning horizons range from four weeks to three months. Applying the week decomposition is not the best choice in this case. First of all, week decomposition requires the optimization to start from the first week and to iteratively solve the rest of the weeks. This would result in a biased calculation toward the first several weeks and reduce the quality of the remaining weeks in the planning horizon. As the algorithm proceeds to the SA block, the last part of the horizon is selected more than the first part because the random selection accounts for the cost of the days in the schedule as discussed earlier. Using nurse decomposition for these larger problems remedies this issue. Assigning nurses by preferences provides more-randomly distributed schedules because the main cost component for a schedule is the violation of cover needs. As a result of choosing the nurse decomposition for these instances, the gap values from the Final Solution increase and they range from 80 to 95 percent.

Table 7 presents a comparison of the studies in the literature applied to the instances to date. Curtois and Qu (2014) report results of an ejection chain heuristic, Gurobi, and B&P. Rahimian et al., 2017b publish findings of an IP&VNS hybrid.

The comparison of the algorithms is based on a time limit on the central processing unit (CPU). Experiments are completed in 10–60 min runtimes. New best-known results reported in the study are marked in

Table 7
Comparison between the studies in the literature.

	10-minute runtime				60-minute runtime				
	IP&VNS Hybrid [Rahimian et al., 2017b]	Ejection Chain [Curtois & Qu, 2014]	Gurobi [Curtois & Qu, 2014]	Our Study	IP&VNS Hybrid [Rahimian et al., 2017b]	Ejection Chain [Curtois & Qu, 2014]	Gurobi [Curtois & Qu, 2014]	B&P [Curtois & Qu, 2014]	Our Study
Instance 01	607	607	607	607	607	607	607	607	607
Instance 02	828	923	828	828	828	837	828	828	828
Instance 03	1001	1003	1001	1001	1001	1003	1001	1001	1001
Instance 04	1716	1719	1716	1716	1716	1718	1716	1716	1716
Instance 05	1143	1439	1143	1143	1143	1358	1143	1160	1143
Instance 06	1950	2344	1950	1950	1950	2258	1950	1952	1950
Instance 07	1056	1284	1056	1056	1056	1269	1056	1058	1056
Instance 08	1364	2529	8995	1341	1344	2260	1323	1308	1322
Instance 09	439	474	439	439	439	463	439	439	439
Instance 10	4631	4999	4631	4631	4631	4797	4631	4631	4631
Instance 11	3443	3967	3443	3443	3443	3661	3443	3443	3443
Instance 12	4042	5611	4045	4044	4040	5211	4040	4046	4040
Instance 13	3109	8707	500410	3200	1905	3037	3109	–	2900
Instance 14	1281	2542	1482	1295	1279	1847	1280	–	1280
Instance 15	4144	6049	78144	4420	3928	5935	4964	–	4190
Instance 16	3306	4343	3521	3253	3225	4048	3233	3323	3225
Instance 17	5760	7835	6149	6138	5750	7835	5851	–	5848
Instance 18	5049	6404	7950	5000	4662	6404	4760	–	4650
Instance 19	3974	6522	29968	3809	3224	5531	5420	–	3218

bold print. In general, the proposed hybrid is very efficient. It achieves optimality for smaller problems and performs better than the EC, Gurobi, and B&P optimizations in most cases. The proposed solution also outperforms the powerful and efficient IP&VNS Hybrid in seven test instances.

The solutions of the hybrid algorithm presented in the study are also available along with additional NRP materials on the following website <https://github.com/ORProjects/NRP>.

5. Conclusion

The efficient usage of scarce resources, especially nurses, in health care is an extremely important task. Yet many scheduling departments and head nurses still create schedules manually. Automated scheduling solutions will improve the utilization of and fairness among nurses. The NRP problem addresses this critical need by automating the assignment of nurses to shifts and days according to hospital needs and nurses' preferences.

This study proposes a hybrid solution for the NRP by integrating MIP-based heuristics and SA. The F&R heuristic generates an initial solution utilizing nurse or week decompositions and feeds it to the SA algorithm, which then applies a variety of neighborhoods to improve the solution. When a solution can no longer be improved, it is routed to another MIP-based heuristic. The F&O heuristic freezes low-cost days and optimizes only the remaining days. Integrating the F&O algorithm into the SA algorithm has several benefits. First of all, the IP solver generally provides better solutions, and therefore intensifies the search. Sometimes, the F&O may not provide better solutions due to a time limit or other parameters, but it still diversifies the search space, which leads to better performance of the neighborhoods in the SA algorithm.

This hybrid framework is tested using the publicly available problems. Seven new best-known results are reported, and the results are compared to the state-of-the-art solution techniques in the literature. The algorithm outperforms most of the solution techniques.

Extending the NRP model with couple of real-world scenarios such as incorporations of skills could be a promising future research direction. Another interesting experiment could be to investigate the performance of mat-heuristics on the NRP instances as this technique is gaining a momentum. Additionally, testing the proposed hybrid framework on other personnel scheduling problems such as high school timetabling could yield valuable results.

CRedit authorship contribution statement

Aykut Melih Turhan: Methodology, Software, Validation, Formal analysis, Investigation, Writing - original draft. **Bilge Bilgen:** Conceptualization, Methodology, Writing - review & editing, Supervision.

Acknowledgements

Data gathering is a time consuming task. When it comes to health care, it is especially difficult, as personal health data is very sensitive. For this reason, we are thankful to Curtois and Qu (2014) for providing the datasets, explanations, and related links to proper sites in a single website.

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